

Student's name

Student's number

Teacher's name



PLC PRESBYTERIAN
LADIES' COLLEGE
SYDNEY
1888

2014
TRIAL
HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics Extension 2

General Instructions

- Reading time - 5 minutes
- Working time - 3 hours
- Write using blue or black pen
Black is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total Marks – 100

Section I: Pages 3-6 10 marks

- Attempt questions 1-10, using the answer sheet on page 19.
- Allow about 15 minutes for this section

Section II: Pages 7-16 90 marks

- Attempt questions 11-16, using the booklets provided.
- Allow about 2 hours 45 minutes for this section

Multiple Choice	11	12	13	14	15	16	Total
							%

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Section I

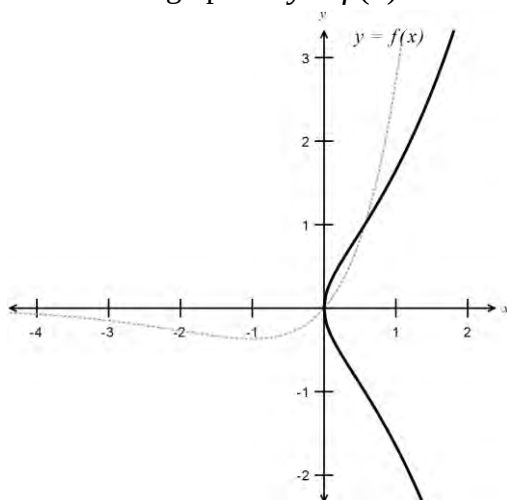
10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1. If $z = (1 - i\sqrt{3})^{2014}$ what is $\text{Arg } z$?
- (A) $-\frac{2014\pi}{3}$
- (B) $-\frac{2014\pi}{6}$
- (C) $\frac{2014\pi}{3}$
- (D) $\frac{2014\pi}{6}$
2. What does the equation $x^2 + 2y^2 - 24 = 0$ represent?
- (A) Parabola
- (B) Hyperbola
- (C) Ellipse
- (D) None of these
3. Which of the following transformations best describes the graph below? The graph of $y = f(x)$ is shown on the same diagram.

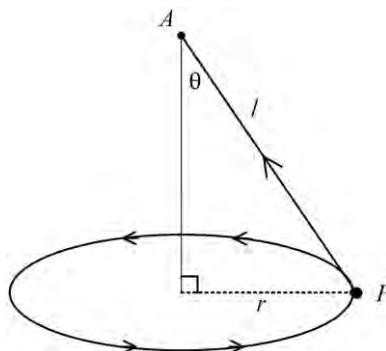


- (A) $|y| = f(x)$
- (B) $y^2 = f(x)$
- (C) $y = |f(x)|$
- (D) $y = [f(x)]^2$

4. Which expression is equal to $\int \frac{dx}{\sqrt{4x^2+1}}$?

- (A) $\sin^{-1} 2x + c$
- (B) $\log_e \left(2x + \sqrt{4x^2+1} \right) + c$
- (C) $\frac{1}{2} \log_e \left(x + \sqrt{x^2 + \frac{1}{4}} \right) + c$
- (D) $\frac{1}{4x} \sqrt{4x^2+1} + c$

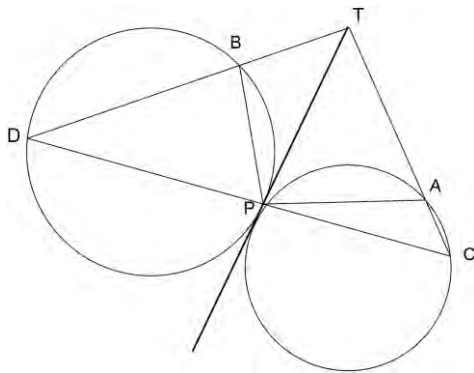
5. A particle, P , of mass m kilograms, is suspended from a fixed point by a string of length, l metres with acceleration due to gravity, $g \text{ ms}^{-2}$. P is moving with uniform circular motion about a horizontal circle with velocity $\omega \text{ rads / second}$ and radius r . The forces acting on the particle are the gravitational force and the tension force T along the string. 1



Which of the following expressions are the correct horizontal and vertical components of the force acting on P ?

- (A) $T \sin \theta = mg$
 $T \cos \theta = mr\omega$
- (B) $T \cos \theta = mg$
 $T \sin \theta = mr\omega$
- (C) $T \sin \theta - mg = 0$
 $T \cos \theta = mr\omega^2$
- (D) $T \cos \theta - mg = 0$
 $T \sin \theta = mr\omega^2$

6. If TP is a common tangent to the circles in the diagram below, which line has an error in proving that $ATBP$ is a cyclic quadrilateral?

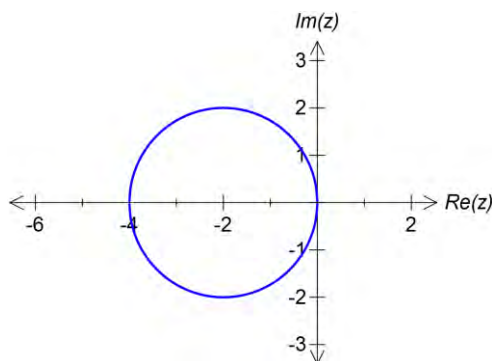


- (A) $\angle TPA = \angle TPB$ (common tangent bisects $\angle APB$)
 (B) $\angle TPA = \angle PCA$ (angle between the tangent and the chord is equal to the angle in the alternate segment)
 (C) $\angle TPB = \angle PDB$ (angle between the tangent and the chord is equal to the angle in the alternate segment)
 (D) $\angle DTC = 180 - \angle TDC - \angle TCD$ (angle sum of a triangle)
 $\therefore \angle APB + \angle DTC = 180$
 Opposite angles in a cyclic quadrilateral are supplementary
 $\therefore ATBP$ is a cyclic quadrilateral

7. A particle is moving in a circular path of radius r , with a constant angular speed of ω . The normal component of the acceleration is:

- (A) ω
 (B) $r\omega$
 (C) $r\omega^2$
 (D) $(r\omega)^2$

8.



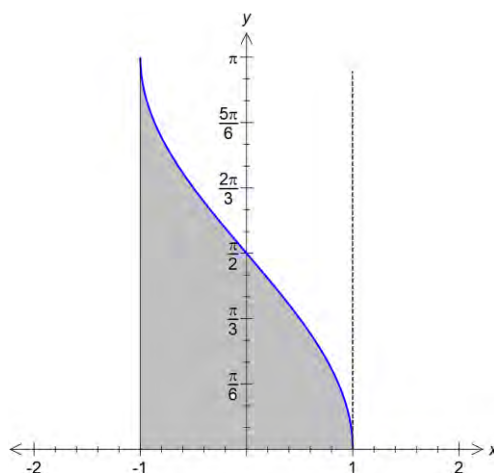
Which one of the following is the equation of the circle in the diagram above?

- (A) $(z + 2)(\bar{z} + 2) = 4$
 (B) $(z - 2)(\bar{z} + 2) = 4$
 (C) $(z - 2)(\bar{z} - 2) = 4$
 (D) $(z + 2i)(\bar{z} - 2i) = 4$

9. The roots of $x^3 + 5x + 3 = 0$ are α , β and γ . Which one of the following polynomials has roots $\alpha\beta$, $\beta\gamma$ and $\alpha\gamma$?

- (A) $x^3 - 5x^2 - 9 = 0$
 (B) $x^3 + 5x^2 + 9 = 0$
 (C) $x^3 - 125x - 375 = 0$
 (D) $x^3 + 125x - 375 = 0$

10. In the diagram, the shaded region is bounded by the x -axis, the line $x = -1$ and the curve $y = \cos^{-1} x$.



Find the volume of the solid formed when this region is rotated about $x = 1$.

- (A) $\frac{3 + \pi^2}{2}$
 (B) $\frac{3}{2}$
 (C) $\frac{5\pi^2}{2}$
 (D) None of the above

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

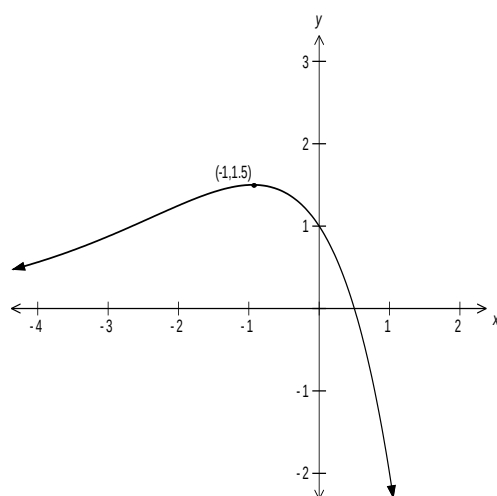
Question 11 (15 marks) Use a SEPARATE writing booklet.

- a) (i) Show that $\tan^3 x = \sec^2 x \tan x - \tan x$. 1
- (ii) Hence evaluate $\int_0^{\frac{\pi}{4}} \tan^3 x \, dx$ 2
- b) If $\omega = \frac{1-i\sqrt{3}}{2}$
- (i) Show that $\omega^3 = -1$. 2
- (ii) Hence calculate ω^{16} 1
- c) (i) Find $\sqrt{5-12i}$ in $x+iy$ form. 2
- (ii) Hence, or otherwise, solve the equation $z^2 + 4z - 1 + 12i = 0$ 2
- d) Consider the equation $z^3 - z^2 - 2z - 12 = 0$. Given that $z = 2\text{cis}\left(\frac{2\pi}{3}\right)$ is a root of the equation, factorise fully over the
- (i) real field 2
- (ii) complex field 1

Question 11 continued over page

Question 11 continued

- e) The following diagram shows the graph of $y = f(x)$.



On your answer sheet, draw separate one-third page sketches of the graphs of the following

- | | | |
|------|-------------------|----------|
| (i) | $y = -f(x)$ | 1 |
| (ii) | $y = \sqrt{f(x)}$ | 1 |

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

- a)** The equation of the ellipse, E , is $\frac{x^2}{25} + \frac{y^2}{9} = 1$.

The point P is on the ellipse with co-ordinates (x_1, y_1) .

- (i) Find the eccentricity of the ellipse. **1**
- (ii) Find the co-ordinates of the foci and the equations of the directrices of the ellipse. **2**
- (iii) Show that the equation of the tangent at P is $\frac{x_1 x}{25} + \frac{y_1 y}{9} = 1$. **2**
- (iv) Let the tangent at P meet a directrix at a point J . Show that $\angle PSJ$ is a right angle where S is the corresponding focus. **3**

- b)** Consider $f(x) = \sin x + \cos x$

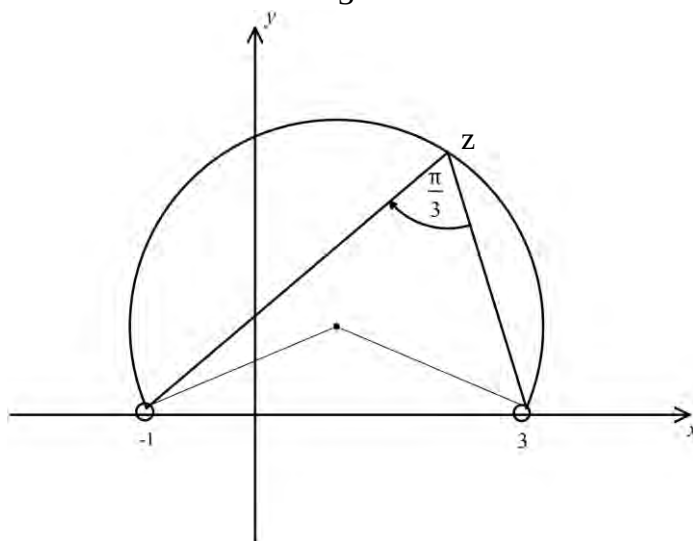
- (i) Find A and B such that $\sin x + \cos x = A \sin(x + B)$ **2**
- (ii) Sketch $f(x) = \sin x + \cos x$ for $-2\pi \leq x \leq 2\pi$. **2**
- (iii) Hence, or otherwise, sketch $y = \frac{1}{f(x)}$ for $-2\pi \leq x \leq 2\pi$. **1**
- (iv) Sketch $y = \frac{f(x)}{x}$ **2**

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

- a) The diagram shows the locus of a point z in the complex plane such that 3

$$\arg(z-3) - \arg(z+1) = \frac{\pi}{3}.$$



This locus is part of a circle. The angle between the lines from -1 to z and from 3 to z is $\frac{\pi}{3}$, as shown.

Find the centre and radius of the circle.

- b) Find $\int \frac{x^2 + 2x}{(x-2)(x^2 + 4)} dx$ 3

- c) Evaluate $\int_0^{\frac{\pi}{4}} \frac{dx}{\cos^2 x + 2 \sin^2 x}$ using $t = \tan x$ 4

- d) (i) Show that $\frac{1 + \sin \theta + i \cos \theta}{1 + \sin \theta - i \cos \theta} = \sin \theta + i \cos \theta$ 3

- (ii) Hence prove that 2
- $$\left(\frac{1 + \sin \theta + i \cos \theta}{1 + \sin \theta - i \cos \theta} \right)^n = \cos \left(\frac{n\pi}{2} - n\theta \right) + i \sin \left(\frac{n\pi}{2} - n\theta \right), \text{ where } n \text{ is a positive integer.}$$

End of Question 13

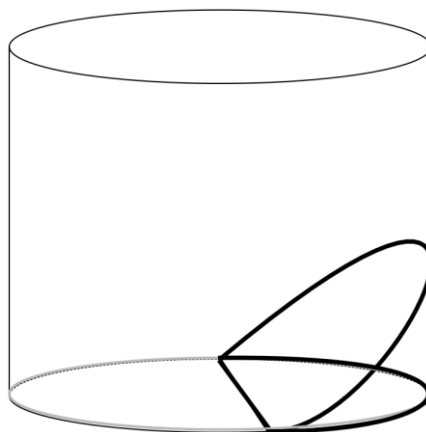
Question 14 (15 marks) Use a SEPARATE writing booklet.

- a) Find all the roots of the equation $16x^3 - 4x^2 - 8x + p = 0$ if two of the roots are equal. **3**
- b) Prove by Mathematical Induction that $2^n > n^2$ for all integers $n \geq 5$ **3**
- c) A particle of mass, m kilograms, is initially projected from the ground at an angle of θ , to the horizontal where $\theta = \tan^{-1}\left(\frac{3}{4}\right)$ and an initial velocity of 25 m/s .
- The equations of motion for the particle are
- $$x = 25t \cos \theta$$
- $$y = -\frac{1}{2}gt^2 + 25t \sin \theta, \text{ where } g \text{ is the acceleration due to gravity.}$$
- (DO NOT PROVE THESE RESULTS)
- (i) Show that $x = 20t$ and $y = 15t - 5t^2$ if $g = 10\text{ m/s}^2$. **2**
- (ii) If the particle hits a wall 40 metres from the point of projection
- (α) find the height above the ground the particle hits. **1**
- (β) show that the velocity of the particle, at the point of impact, is $\sqrt{425}\text{ m/s}$. **2**
- (iii) At impact, the particle is instantaneously at rest. It then falls vertically to the ground with a resistance force acting against the vertical motion equal to $0.01mv^2$ Newtons.
- (α) Show that $a = 10 - 0.01v^2$, where a is the acceleration and v is the velocity of the particle. **1**
- (β) Find the velocity on returning to the ground. Answer correct to 2 decimal places. **3**

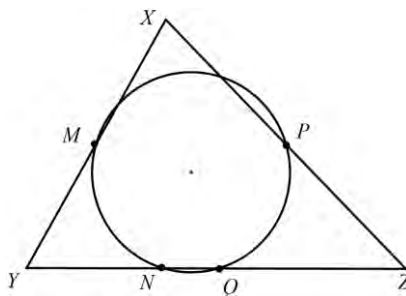
End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

- a) A wedge is cut out of a right circular cylinder of radius 4 centimetres by two planes. One plane is perpendicular to the axis of the cylinder. The other plane intersects the first at an angle of 30° , along a diameter of the cylinder. Find the volume of the wedge. 3



- b) In the acute-angled triangle XYZ , M is the midpoint of XY , Q is the midpoint of YZ and P is the midpoint of ZX . The circle through M , Q and P also cuts YZ at N as shown in the diagram.

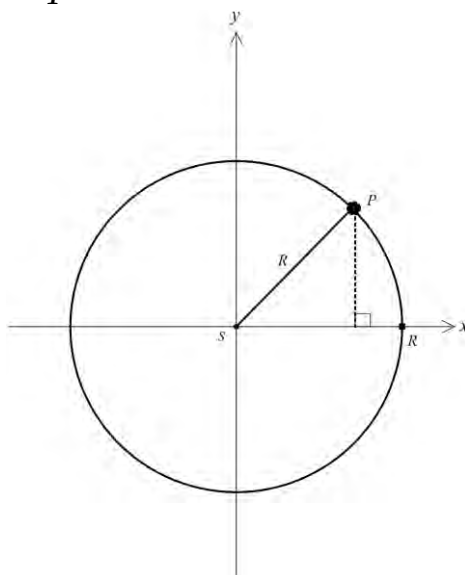


- | | | |
|-------|-----------------------------------|---|
| (i) | Prove $MPQY$ is a parallelogram. | 1 |
| (ii) | Prove $\angle MNY = \angle MPQ$. | 1 |
| (iii) | Prove that $XN \perp YZ$. | 2 |

Question 15 continued over page

Question 15 continued

- c) A planet P of mass, m kilograms, moves in a circular orbit of radius R metres, around a star, S , in uniform circular motion. The position of the planet at time t seconds is given by the equations $x = R \cos \frac{2\pi t}{T}$ and $y = R \sin \frac{2\pi t}{T}$, where T is a constant.



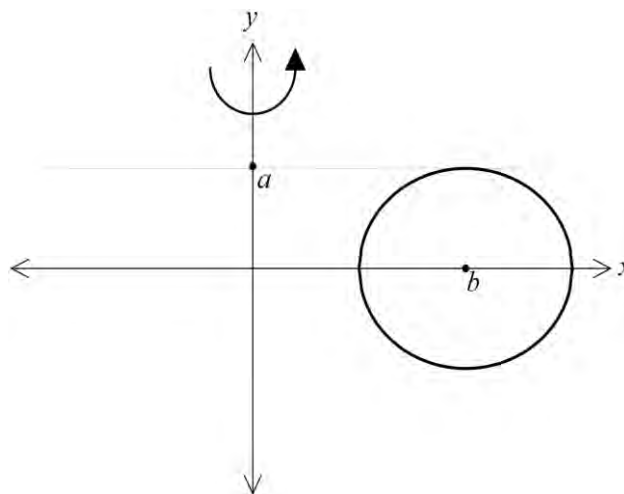
- | | | |
|-------|---|---|
| (i) | Show that $\ddot{x} = -\frac{4\pi^2}{T^2}x$ and $\ddot{y} = -\frac{4\pi^2}{T^2}y$ | 2 |
| (ii) | Show the acceleration of P is $-\frac{4\pi^2}{T^2}R$. | 1 |
| (iii) | Find the force exerted by the star, S , on the planet, P . | 1 |
| (iv) | It is known that the magnitude of the gravitational force pulling the planet towards the star is given by $F = \frac{GMm}{R^2}$, where G is constant and M is the mass of the star, S , in kilograms. Show that the expression for T in terms of R , M and G is $T = 2\pi R \sqrt{\frac{R}{GM}}$. | 2 |

Question 15 continued over page

Question 15 continued

- d)** A donut shaped solid called a torus is formed by revolving $(x-b)^2 + y^2 = a^2, 0 < a < b$ about the y -axis.

2



Express the volume of the torus as a definite integral in x . Do not evaluate this integral.

End of Question 15

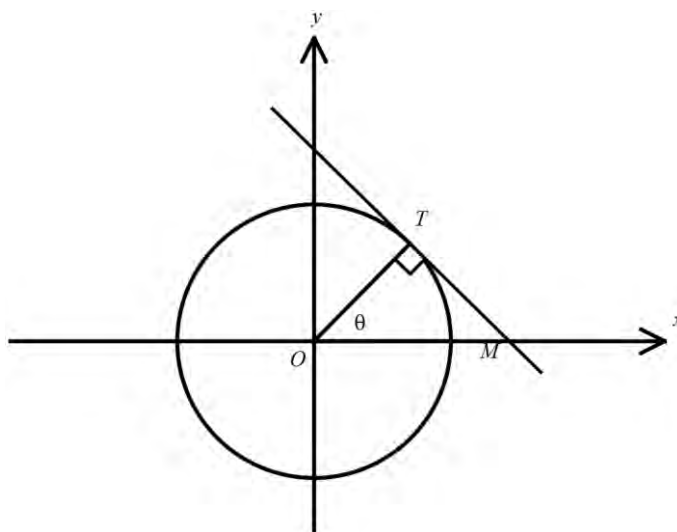
Question 16 (15 marks) Use a SEPARATE writing booklet.

a) If $I_n = \int_0^{\frac{\pi}{2}} \sin^n \theta d\theta$ where $n \geq 2$,

(i) Show that $I_n = \frac{(n-1)}{n} I_{n-2}$ **3**

(ii) Hence or otherwise, evaluate $\int_0^2 (4-x^2)^{\frac{5}{2}} dx$. **3**

b) The figure shows the circle $x^2 + y^2 = a^2$.



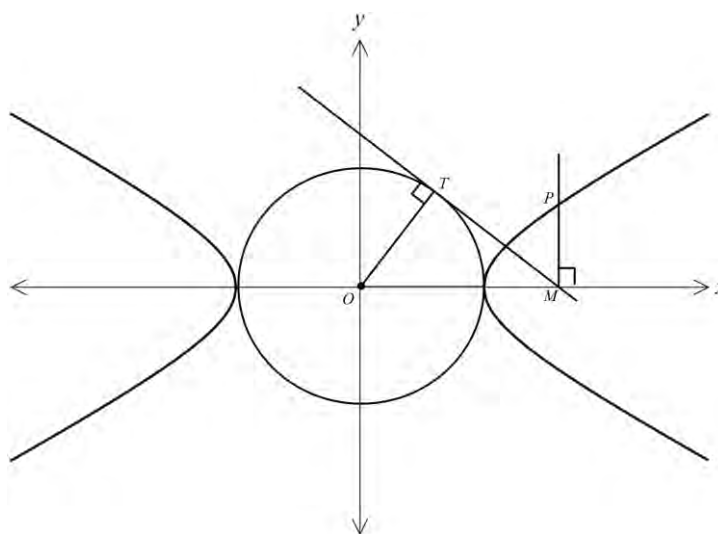
The point T lies on the circle. $\angle TOx = \theta$, where $0 \leq \theta \leq \frac{\pi}{2}$. The tangent to the circle at T meets the x -axis at M .

(i) Show that the co-ordinates of M are $(a \sec \theta, 0)$. **1**

Question 16 continued over page

Question 16 continued

The hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and the circle $x^2 + y^2 = a^2$ where $a, b > 0$ are shown on the diagram below:



MP is perpendicular to Ox and P is a point on the hyperbola in the first quadrant.

- (ii) Show that the co-ordinates of P are $(a \sec \theta, b \tan \theta)$. 2
- (iii) If Q is another point on the hyperbola with co-ordinates $(a \sec \phi, b \tan \phi)$ where $\theta + \phi = \frac{\pi}{2}$ and $\theta \neq \frac{\pi}{4}$, show that the equation of the chord PQ is $y = \frac{b}{a}(\cos \theta + \sin \theta)x - b$. 3
- (iv) Show that every such chord passes through a fixed point and determine its co-ordinates. 1
- (v) State the equation of the asymptotes for the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. 1
- (vi) Show that as $\theta \rightarrow \frac{\pi}{2}$, the chord PQ approaches a line parallel to an asymptote. 1

End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

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Extension 2 Mathematics

Multiple Choice Answer Sheet

Student Number_____

Completely fill the response oval representing the most correct answer.

- | | | | | | | | | |
|-----|---|-----------------------|---|-----------------------|---|-----------------------|---|-----------------------|
| 1. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 2. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 3. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 4. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 5. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 6. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 7. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 8. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 9. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 10. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |

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Solutions for exams and assessment tasks

Academic Year	Yr 12	Calendar Year	2014
Course	Ext. 2	Name of task/exam	Trials

1. $z = (1 - i\sqrt{3})^{2014}$

$$\text{Arg}(1 - i\sqrt{3}) = \tan^{-1} \frac{-\sqrt{3}}{1} = -\frac{\pi}{3}$$

$$\therefore z = (1 - i\sqrt{3})^{2014}$$

$$\text{Arg}(1 - i\sqrt{3})^{2014} = 2014 \text{ Arg}(1 - i\sqrt{3})$$

$$= 2014 \left(-\frac{\pi}{3}\right)$$

$$= -\frac{2014\pi}{3}$$

$\therefore A$

2. $x^2 + 2y^2 - 24 = 0$

$$x^2 + 2y^2 = 24$$

$$\frac{x^2}{24} + \frac{y^2}{12} = 1$$

$\therefore$ ellipse

$\therefore C$

3. B

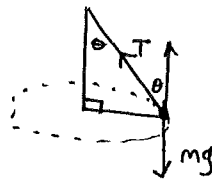
4. $\int \frac{dx}{\sqrt{4x^2 + 1}}$

$$= \int \frac{dx}{2\sqrt{x^2 + (\frac{1}{2})^2}}$$

$$= \frac{1}{2} \ln \left(x + \sqrt{x^2 + \frac{1}{4}} \right) + c$$

$\therefore C$

5.



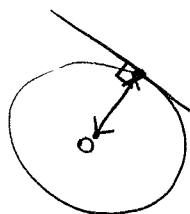
Horizontally: $T \sin \theta = mr\omega^2$

Vertically: $T \cos \theta = mg$

$\therefore D$

6. A

7.



normal component is the F acting towards the centre of the circle

$$\therefore a = r\omega^2$$

$\therefore C$

8. Circle in diagram is

$$|z + 2| = 2$$

$$\therefore \sqrt{(x+2)^2 + y^2} = 2$$

$$(x+2)^2 + y^2 = 4$$

Consider $(z+2)(\bar{z}+2) = 4$

$$4 = (x+iy+2)(x-iy+2)$$

$$4 = x^2 + y^2 + 2(x+iy) + 2(x-iy) + 4$$

$$0 = x^2 + y^2 + 4x$$

$$0 = (x+2)^2 + y^2 - 4$$

$\therefore A$

Academic Year		Calendar Year	
Course		Name of task/exam	

9. $x^3 + 5x + 3 = 0$

α, β, γ

$\alpha + \beta + \gamma = -\frac{b}{a} = 0 \quad (1)$

$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} = 5 \quad (2)$

$\alpha\beta\gamma = -\frac{d}{a} = -3 \quad (3)$

If roots $\alpha\beta, \beta\gamma, \alpha\gamma$

sum of roots 1 at a time:

$\alpha\beta + \beta\gamma + \alpha\gamma = 5 \quad \text{from (2)}$

sum of roots 2 at a time:

$\alpha\beta\gamma + \alpha\beta\alpha\gamma + \beta\gamma\alpha\gamma$

$= \alpha\beta\gamma (\beta + \alpha + \gamma)$

$= -3(0)$

$= 0$

product of roots

$\alpha\beta\gamma\alpha\gamma = \alpha^2\beta^2\gamma^2$

$= (\alpha\beta\gamma)^2$

$= (-3)^2$

$= 9$

$\therefore$ polynomial is

$x^3 - (\alpha\beta + \beta\gamma + \alpha\gamma)x^2 + (\alpha\beta\gamma(\alpha + \beta + \gamma))x - \alpha^2\beta^2\gamma^2 = 0$

$x^3 - 5x^2 + 0x - 9 = 0$

$x^3 - 5x^2 - 9 = 0$

$\therefore A$

10. $A(y) = \pi(R^2 - r^2)$

$= \pi(R - r)(R + r)$

$R = 2$

$r = 1 - x$

$\therefore A(y) = \pi(2 - 1 + x)(2 + 1 - x)$

$= \pi(1 + x)(3 - x)$

$V = \pi \int_0^\pi (1 + \cos y)(3 - \cos y) dy$

Note $y = \cos^{-1}x$
 $\cos y = x$

$\therefore V = \pi \int_0^\pi (3 + 2\cos y - \cos^2 y) dy$

$= \pi \left[[3y + 2\sin y]_0^\pi - \int_0^\pi \cos^2 y dy \right]$

$= \pi \left[(3\pi + 0) - 0 - \int_0^\pi \left(\frac{1}{2} \cos 2y + \frac{1}{2} \right) dy \right]$

$= \pi \left[3\pi - \left[\frac{1}{2}y + \frac{1}{4} \sin 2y \right]_0^\pi \right]$

$= \pi \left[3\pi - \left(\frac{\pi}{2} - 0 - 0 \right) \right]$

$= 3\pi^2 - \frac{\pi^2}{2}$

$= \frac{5\pi^2}{2}$

$\therefore C$

Academic Year		Calendar Year	
Course		Name of task/exam	

Q11

a) i) RTS $\tan^3 x = \sec^2 x \tan x - \tan x$

$$\text{RHS} = \tan x (\sec^2 x - 1)$$

$$= \tan x (\tan^2 x)$$

$$= \tan^3 x$$

$$= \text{LHS}$$

ii) $\int_0^{\pi/4} \tan^3 x \, dx$

$$= \int_0^{\pi/4} (\sec^2 x \tan x - \tan x) \, dx$$

$$= \left[\frac{\tan^2 x}{2} \right]_0^{\pi/4} - \int_0^{\pi/4} \frac{\sin x}{\cos x} \, dx$$

$$= \left(\frac{1}{2} - 0 \right) + \left[\ln(\cos x) \right]_0^{\pi/4}$$

$$= \frac{1}{2} + (\ln \frac{1}{\sqrt{2}} - \ln 1)$$

$$= \frac{1}{2} + \ln \frac{1}{\sqrt{2}}$$

$$= \frac{1}{2} + \ln 2^{-1/2}$$

$$= \frac{1}{2} - \frac{1}{2} \ln 2.$$

b) $\omega = \frac{1 - \sqrt{3}i}{2}$

i) $\omega^3 = \left(\frac{1 - \sqrt{3}i}{2} \right)^3$

$$= \left(\frac{1 - 2\sqrt{3}i - 3}{4} \right) \left(\frac{1 - \sqrt{3}i}{2} \right)$$

$$= \left(\frac{-2 - 2\sqrt{3}i}{4} \right) \left(\frac{1 - \sqrt{3}i}{2} \right)$$

$$\omega^3 = \frac{-2 + 2\sqrt{3}i - 2\sqrt{3}i - 6}{8}$$

$$= -\frac{8}{8}$$

$$= -1.$$

$$\therefore \omega^3 = -1$$

ii) $\omega^{16} = (\omega^3)^5 \times \omega$

$$= (-1)^5 \omega$$

$$= -\omega$$

OR $\frac{-1 + \sqrt{3}i}{2}$

c) i) $\sqrt{5 - 12i} = a + ib$

$$5 - 12i = (a + ib)^2$$

$$5 - 12i = a^2 - b^2 + 2abi$$

equating:

$$5 = a^2 - b^2$$

$$-12 = 2ab$$

$$b = -\frac{6}{a}$$

$$\therefore 5 = a^2 - \left(-\frac{6}{a} \right)^2$$

$$5 = a^2 - \frac{36}{a^2}$$

$$5a^2 = a^4 - 36$$

$$(a^2 + 4)(a^2 - 9) = 0$$

$$a = \pm 3, a \text{ real}$$

$$\therefore b = \mp 2$$

$$\therefore \sqrt{5 - 12i} = \pm (3 - 2i)$$

Solutions for exams and assessment tasks

Academic Year		Calendar Year	
Course		Name of task/exam	

$$\text{ii } z^2 + 4z - 1 + 12i = 0$$

$$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$z = \frac{-4 \pm \sqrt{16 - 4(-1 + 12i)}}{2}$$

$$= \frac{-4 \pm \sqrt{16 + 4 - 48i}}{2}$$

$$= \frac{-4 \pm \sqrt{20 - 48i}}{2}$$

$$= \frac{-4 \pm 2\sqrt{5 - 12i}}{2}$$

$$= -2 \pm \sqrt{5 - 12i}$$

$$= -2 \pm (3 - 2i)$$

$$= -2 + 3 - 2i, -2 - 3 + 2i$$

$$= 1 - 2i, -5 + 2i$$

$$\text{d } z^3 - z^2 - 2z - 12 = 0$$

Since coefficients are real
roots occur in conjugate pairs

$$\therefore 2 \operatorname{cis}\left(\frac{2\pi}{3}\right) = 2\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)$$

$$= -1 + \sqrt{3}i$$

$\therefore -1 - \sqrt{3}i$ is also a
root.

$$\therefore (z - (-1 + \sqrt{3}i))(z - (-1 - \sqrt{3}i))$$

$$= z^2 - z(-1 - \sqrt{3}i) - z(-1 + \sqrt{3}i) + (-1 + \sqrt{3}i)(-1 - \sqrt{3}i)$$

$$= z^2 + z + \sqrt{3}iz + z - \sqrt{3}iz + 1 + 3$$

$$= z^2 + 2z + 4$$

$$\therefore z^2 + 2z + 4 \quad \begin{array}{r} z + 3 \\ z^3 - z^2 - 2z - 12 \\ \hline z^3 + 2z^2 + 4z \\ \hline -3z^2 - 6z - 12 \\ \hline -3z^2 - 6z - 12 \\ \hline \end{array}$$

$\therefore i$ real field

$$(z^2 + 2z + 4)(z - 3)$$

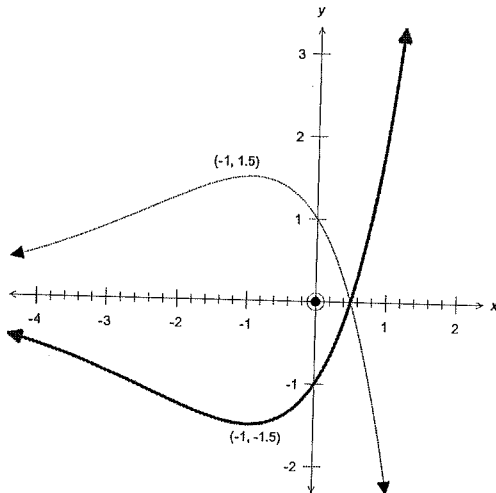
ii complex field

$$(z - (-1 + \sqrt{3}i))(z - (-1 - \sqrt{3}i))(z - 3)$$

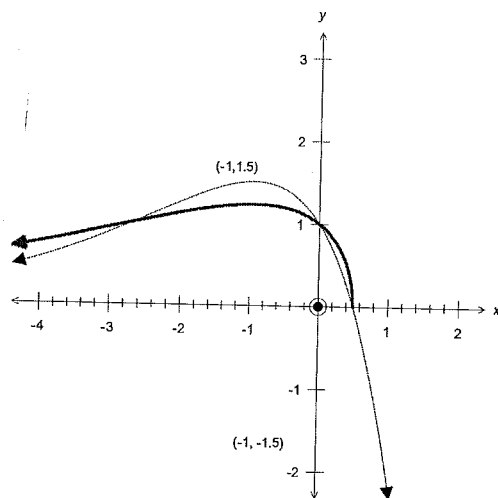
e, see separate sheet.

Academic Year		Calendar Year	
Course		Name of task/exam	

11e i)



11e ii)



Academic Year		Calendar Year	
Course		Name of task/exam	

Q12

$$a. \quad \frac{x^2}{25} + \frac{y^2}{9} = 1$$

$$\begin{aligned} \therefore e^2 &= 1 - \left(\frac{b}{a}\right)^2 \\ &= 1 - \left(\frac{3}{5}\right)^2 \\ e &= \sqrt{\frac{16}{25}} \quad e > 0 \end{aligned}$$

$$e = \frac{4}{5}$$

$$\begin{aligned} \text{ii foci } (\pm ae, 0) \\ &= (\pm 4, 0) \end{aligned}$$

directrices

$$x = \pm \frac{a}{e}$$

$$x = \pm \frac{25}{4}$$

$$\text{iii} \quad \frac{x^2}{25} + \frac{y^2}{9} = 1$$

$$\frac{2x}{25} + \frac{2y}{9} \frac{dy}{dx} = 0$$

$$\frac{2y}{9} \frac{dy}{dx} = -\frac{2x}{25}$$

$$\frac{dy}{dx} = -\frac{x}{25} \times \frac{9}{y}$$

at (x_1, y_1)

$$m = -\frac{9x_1}{25y_1}$$

 $\therefore$ eqn tangent:

$$y - y_1 = -\frac{9x_1}{25y_1} (x - x_1)$$

$$25yy_1 - 25y_1^2 = -9xx_1 + 9x_1^2$$

$$9xx_1 + 25yy_1 = 9x_1^2 + 25y_1^2$$

$$\div 25$$

$$\frac{xx_1}{25} + \frac{yy_1}{9} = \frac{x_1^2}{25} + \frac{y_1^2}{9}$$

 (x_1, y_1) lies on the ellipse

$$\therefore \frac{x_1^2}{25} + \frac{y_1^2}{9} = 1$$

$$\therefore \frac{xx_1}{25} + \frac{yy_1}{9} = 1$$

iv If tangent meets directrix
then $x = \frac{25}{4}$, $y = ?$

$$\frac{\frac{25}{4}x_1}{25} + \frac{yy_1}{9} = 1$$

$$\frac{x_1}{4} + \frac{yy_1}{9} = 1$$

$$9x_1 + 4yy_1 = 36$$

$$4yy_1 = 36 - 9x_1$$

$$y = \frac{36 - 9x_1}{4y_1}$$

Solutions for exams and assessment tasks

Academic Year		Calendar Year	
Course		Name of task/exam	

$$\therefore J \left(\frac{25}{4}, \frac{36-9x_1}{4y_1} \right) S(4,0)$$

$$m_{ST} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{36-9x_1 - 0}{\frac{25}{4} - 4}$$

$$= \frac{36-9x_1}{4y_1} \div \frac{9}{4}$$

$$= \frac{4(36-9x_1)}{4 \times 9 y_1} = \frac{36(4-x_1)}{36 y_1}$$

$$m_{PS} = \frac{y_1 - 0}{x_1 - 4}$$

$$= \frac{y_1}{x_1 - 4}$$

$$m_{ST} \times m_{PS} = \frac{\cancel{y_1}}{(\cancel{x_1} - 4)} \times \frac{\cancel{4 - x_1}}{\cancel{y_1}}$$

$$= -1$$

$$\text{Since } m_{ST} \times m_{PS} = -1$$

$$ST \perp PS$$

$\therefore \angle PSJ$ is a right angle.

$$b) f(x) = \sin x + \cos x$$

$$\downarrow \sin x + \cos x = A \sin(x+B)$$

$$\sin x + \cos x = A \sin x \cos B + A \cos x \sin B$$

equating:

$$1 = A \cos B \quad (1)$$

$$1 = A \sin B \quad (2)$$

$$1 = \tan B$$

$$B = \frac{\pi}{4}$$

$$1 = A \cos \frac{\pi}{4}$$

$$A = \frac{1}{\frac{1}{\sqrt{2}}}$$

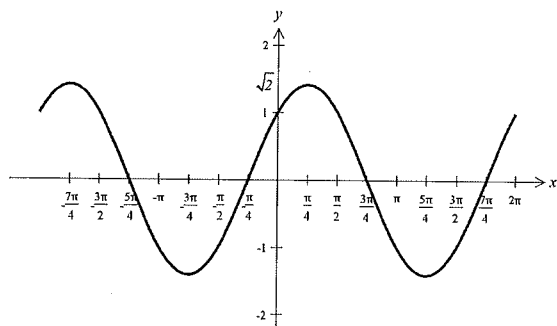
$$= \sqrt{2}$$

$$\therefore \sin x + \cos x = \sqrt{2} \sin \left(x + \frac{\pi}{4} \right)$$

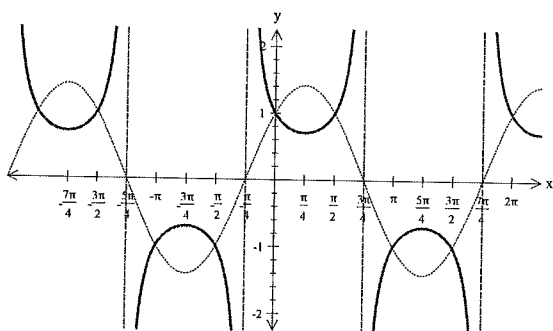
Solutions for exams and assessment tasks

Academic Year		Calendar Year	
Course		Name of task/exam	

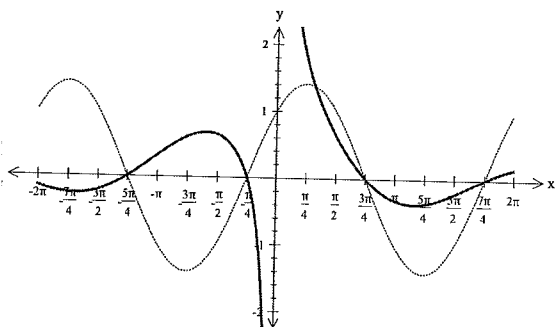
12b ii)



12b iii)



12b iv)

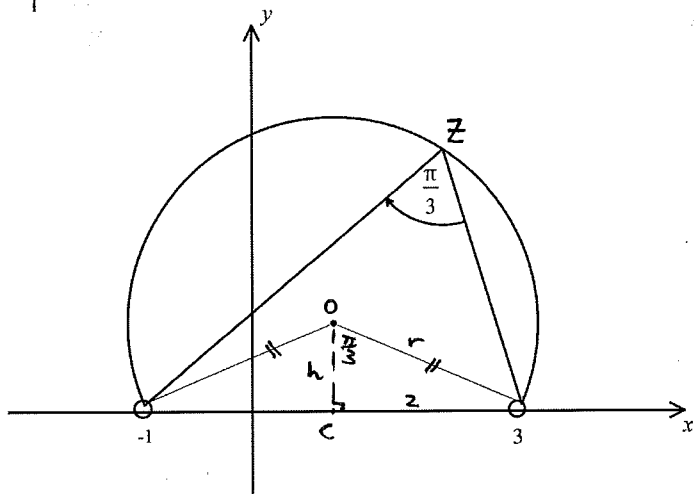


Solutions for exams and assessment tasks

Academic Year		Calendar Year	
Course		Name of task/exam	

Q13

a.



Let O be centre of circle

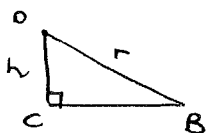
$$A(-1, 0) \quad B(3, 0)$$

$$\angle AOB = \frac{2\pi}{3} \quad (\text{angle centre twice angle at circumference})$$

 $\triangle AOB$ is isosceles

Drop perpendicular from O to real axis. Call this point C.

$$\therefore OC \perp AB$$

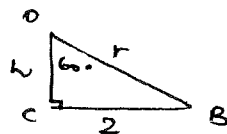


C is midpt AB

$$\therefore C(1, 0)$$

$$\therefore CB = 2 \text{ units.}$$

$$\angle COB = \frac{\pi}{3} \quad (\text{isosceles } \triangle)$$



$$\tan 60 = \frac{2}{h}$$

$$h = \frac{2}{\tan 60} = \frac{2}{\sqrt{3}}$$

$$\sin 60 = \frac{2}{r}$$

$$r = \frac{2}{\sin 60} = \frac{2}{\sqrt{3}/2} = \frac{4}{\sqrt{3}}$$

$$\therefore \text{Centre } \left(1, \frac{2}{\sqrt{3}}\right) \quad R = \frac{4}{\sqrt{3}} \text{ units}$$

$$b \int \frac{x^2 + 2x}{(x-2)(x^2+4)} dx$$

$$\frac{x^2 + 2x}{(x-2)(x^2+4)} = \frac{A}{(x-2)} + \frac{Bx+C}{(x^2+4)}$$

$$x^2 + 2x = A(x^2 + 4) + (Bx + C)(x - 2)$$

$$x^2 + 2x = Ax^2 + 4A + Bx^2 - 2Bx + Cx - 2C$$

equating

$$1 = 4 + B \quad (1)$$

$$2 = -2B + C \quad (2)$$

$$0 = 4A - 2C \quad (3)$$

$$\therefore 4A = 2C$$

$$2A = C$$

$$\therefore 2 = -2B + 2A$$

$$1 = -B + A \quad (2)$$

$$1 = B + A \quad (1)$$

$$2 = 2A \quad (1) + (2)$$

$$A = 1, C = 2, B = 0$$

Academic Year		Calendar Year	
Course		Name of task/exam	

$$\begin{aligned}\therefore \int \frac{x^2 + 2x}{(x-2)(x^2+4)} dx &= \int \left\{ \frac{1}{x-2} + \frac{2}{x^2+4} \right\} dx \\ &= \ln|x-2| + 2 \int \frac{1}{4+x^2} dx \\ &= \ln|x-2| + \frac{2}{2} \tan^{-1} \frac{x}{2} + C\end{aligned}$$

$$\therefore \text{Integral} = \ln|x-2| + \tan^{-1} \frac{x}{2} + C.$$

$$\int_0^{\frac{\pi}{4}} \frac{dx}{\cos^2 x + 2 \sin^2 x} \quad u = \tan x$$

$$= \int_0^{\frac{\pi}{4}} \frac{\frac{dx}{\cos^2 x}}{\frac{\cos^2 x}{\cos^2 x} + \frac{2 \sin^2 x}{\cos^2 x}}$$

$$= \int_0^{\frac{\pi}{4}} \frac{\sec^2 x dx}{1 + 2 \tan^2 x}$$

Given $u = \tan x$

$$du = \sec^2 x dx$$

also when $x = 0$ $u = 0$

when $x = \frac{\pi}{4}$ $u = 1$.

$$\therefore \int_0^1 \frac{du}{1+2u^2}$$

$$= \int_0^1 \frac{du}{2\left(\frac{1}{2} + u^2\right)}$$

$$= \frac{1}{2} \int_0^1 \frac{du}{\frac{1}{2} + u^2}$$

$$= \frac{1}{2} \sqrt{2} \left[\tan^{-1} \frac{u}{\frac{1}{\sqrt{2}}} \right]_0^1$$

$$= \frac{1}{\sqrt{2}} \left(\tan^{-1} \frac{1}{\frac{1}{\sqrt{2}}} - \tan^{-1} 0 \right)$$

$$= \frac{1}{\sqrt{2}} \tan^{-1} \frac{1}{\frac{1}{\sqrt{2}}}$$

$$= \frac{1}{\sqrt{2}} \tan^{-1} \sqrt{2}$$

Academic Year		Calendar Year	
Course		Name of task/exam	

di RTS

$$\frac{1 + \sin \theta + i \cos \theta}{1 + \sin \theta - i \cos \theta} = \sin \theta + i \cos \theta$$

$$\begin{aligned} \text{LHS} &= \frac{1 + \sin \theta + i \cos \theta}{1 + \sin \theta - i \cos \theta} \times \frac{1 + \sin \theta + i \cos \theta}{1 + \sin \theta + i \cos \theta} \\ &= \frac{(1 + \sin \theta)^2 + i \cos \theta (1 + \sin \theta) + i \cos \theta (1 + \sin \theta)}{(1 + \sin \theta)^2 + \cos^2 \theta} \\ &\quad - \cos^2 \theta \\ &= \frac{(1 + \sin \theta)^2 + \cos^2 \theta}{(1 + \sin \theta)^2 + \cos^2 \theta} \end{aligned}$$

$$= \frac{1 + 2\sin\theta + \sin^2\theta + i\cos\theta + i\cos\theta\sin\theta}{(1 + \sin\theta)^2 + \cos^2\theta} + \frac{i\cos\theta + i\cos\theta\sin\theta - \cos^2\theta}{(1 + \sin\theta)^2 + \cos^2\theta}$$

$$= \frac{2 \sin^2 \theta + 2 \sin \theta + 2 i \cos \theta + 2 i \cos \theta \sin \theta}{1 + 2 \sin \theta + \sin^2 \theta + \cos^2 \theta}$$

$$= \frac{2 [\sin^2 \theta + \sin \theta + i \cos \theta + i \cos \theta \sin \theta]}{2 [1 + \sin \theta]}$$

$$= \frac{\sin(\sin\theta + 1) + i\cos\theta(1 + \sin\theta)}{(1 + \sin\theta)}$$

$$= \frac{(1 + \sin \theta)(\sin \theta + i \cos \theta)}{(1 + \sin \theta)}$$

$$= \sin \theta + i \cos \theta$$

$$= \text{R.H.S}$$

ii) RTP

$$\left(\frac{1 + \sin \theta + i \cos \theta}{1 + \sin \theta - i \cos \theta} \right)^2 = \cos \left(\frac{n\pi}{2} - n\theta \right) + i \sin \left(\frac{n\pi}{2} - n\theta \right)$$

$$\text{LHS} = \left(\frac{1 + \sin\theta + i\cos\theta}{1 + \sin\theta - i\cos\theta} \right)^n$$

$$= (\sin \theta + i \cos \theta)^2$$

$$= \left(\cos\left(\frac{\pi}{2} - \theta\right) + i \sin\left(\frac{\pi}{2} - \theta\right) \right)^2$$

$$= \cos \left[n \left(\frac{\pi}{2} - \theta \right) \right] + i \sin \left[n \left(\frac{\pi}{2} - \theta \right) \right]$$

by De Moivre's Thm

$$= \cos\left(\frac{n\pi}{2} - n\theta\right) + i \sin\left(\frac{n\pi}{2} - n\theta\right)$$

$$= R H S$$

∴ proved

Solutions for exams and assessment tasks

Academic Year		Calendar Year	
Course		Name of task/exam	

Q14

a, $P(x) = 16x^3 - 4x^2 - 8x + p = 0$

Let roots be α, α, β

sum roots 1 at time

$$2\alpha + \beta = \frac{4}{16}$$

$$\beta = \frac{1}{4} - 2\alpha$$

Also $P(\alpha) = 0$

$P'(\alpha) = 0$ since double root.

$$\therefore P'(x) = 48x^2 - 8x - 8$$

$$P'(\alpha) = 0$$

$$48\alpha^2 - 8\alpha - 8 = 0$$

$$6\alpha^2 - \alpha - 1 = 0$$

$$(3\alpha + 1)(2\alpha - 1) = 0$$

$$\therefore \left. \begin{array}{l} \alpha = -\frac{1}{3} \\ \alpha = \frac{1}{2} \end{array} \right\} \left. \begin{array}{l} \alpha = \frac{1}{2} \\ \beta = -\frac{3}{4} \end{array} \right\}$$

$\therefore$ roots $-\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}$

and $\frac{1}{2}, \frac{1}{2}, -\frac{3}{4}$

b, Prove by m.I. for $n \geq 5$

$$2^n > n^2$$

Step 1: Prove true for $n=5$

$$\begin{aligned} \text{LHS} &= 2^5 \\ &= 32 \end{aligned}$$

$$\begin{aligned} \text{RHS} &= 5^2 \\ &= 25 \end{aligned}$$

$$\text{LHS} > \text{RHS}$$

$$\therefore \text{true for } n=5$$

Step 2: Assume true for $n=k$

$$\therefore 2^k > k^2$$

Step 3: Prove true for $n=k+1$

i.e. prove

$$2^{k+1} > (k+1)^2$$

$$2^{k+1} = 2^k \cdot 2$$

$$> k^2 \cdot 2 \quad \text{by assumption}$$

$$> 2k^2$$

now $2k^2$ at $k=5$ gives 50.

$(k+1)^2$ at $k=5$ gives 36.

clearly $2k^2 > (k+1)^2$

for all values of $k \geq 5$

$$\therefore 2^{k+1} > 2k^2 > (k+1)^2$$

$$\therefore 2^{k+1} > (k+1)^2$$

$\therefore$ proved by m.I. for all values of $n \geq 5$.

Solutions for exams and assessment tasks

Academic Year		Calendar Year	
Course		Name of task/exam	

14b another method:

Step 3: Prove true for $n = k+1$

i.e. prove $2^{k+1} > (k+1)^2$

If we can prove $2^{k+1} - (k+1)^2 > 0$
then we have proved $2^{k+1} > (k+1)^2$.

$$\text{LHS} = 2^{k+1} - (k+1)^2$$

$$= 2^k \times 2 - (k^2 + 2k + 1)$$

$$> 2k^2 - (k^2 + 2k + 1)$$

by assumption

$$= 2k^2 - k^2 - 2k - 1$$

$$= k^2 - 2k - 1$$

$$= (k-1)^2 - 2$$

$$> 0 \quad \text{if } k > 3.$$

Since we have $k \geq 5$,
this must also be true

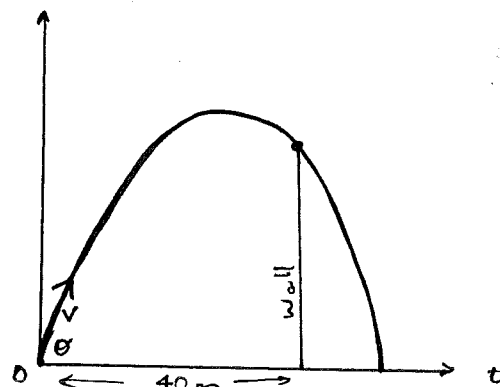
$$\therefore 2^{k+1} - (k+1)^2 > 0$$

$$\therefore 2^{k+1} > (k+1)^2$$

$\therefore$ proved by M.I.

for all values of $n \geq 5$

c

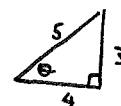


$$g = 10$$

$$x = 25t \cos \theta$$

$$y = -\frac{1}{2} g t^2 + 25t \sin \theta$$

i we know $\tan \theta = \frac{3}{4}$
 $v = 25 \text{ m/s}$



$$x = 25t \left(\frac{4}{5} \right)$$

$$x = 20t$$

$$y = -\frac{1}{2} (10) t^2 + 25t \left(\frac{3}{5} \right)$$

$$y = -5t^2 + 15t$$

ii Δ when $x = 40$

$$40 = 20t$$

$$t = 2.$$

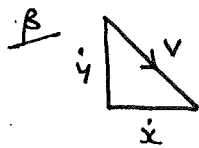
$$\therefore y = -5(2)^2 + 15(2)$$

$$= -20 + 30$$

$$= 10$$

$\therefore$ 10 metres high.

Academic Year		Calendar Year	
Course		Name of task/exam	



if $x = 20t$

$$\dot{x} = 20$$

if $y = 15t - 5t^2$

$$\dot{y} = 15 - 10t$$

at $t = 2$

$$\dot{y} = 15 - 20$$

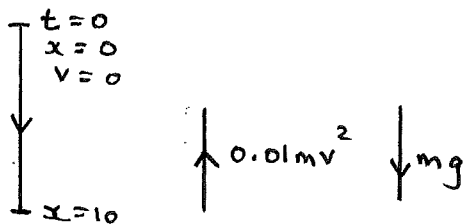
$$\dot{y} = -5$$

$$\therefore v = \sqrt{\dot{x}^2 + \dot{y}^2}$$

$$= \sqrt{20^2 + (-5)^2}$$

$$v = \sqrt{425} \text{ m/s}$$

iii



α $F = ma$

$$ma = mg - 0.01mv^2$$

$$a = g - 0.01v^2$$

$$a = 10 - 0.01v^2$$

β $v \frac{dv}{dx} = 10 - 0.01v^2$

$$\frac{dv}{dx} = \frac{10 - 0.01v^2}{v}$$

$$\frac{dx}{dv} = \frac{v}{10 - 0.01v^2}$$

$$\int_0^{10} dx = \int_0^V \frac{v}{10 - 0.01v^2} dv$$

$$\left[x \right]_0^{10} = \frac{1}{-0.02} \int_0^V \frac{-0.02v}{10 - 0.01v^2} dv$$

$$10 = -\frac{1}{0.02} \left[\ln |10 - 0.01v^2| \right]_0^V$$

$$-0.2 = \ln |10 - 0.01V^2| - \ln |10 - 0|$$

$$-0.2 = \ln \left| \frac{10 - 0.01V^2}{10} \right|$$

$$e^{-0.2} = \frac{10 - 0.01V^2}{10}$$

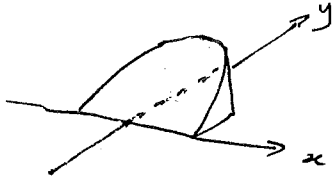
$$10e^{-0.2} = 10 - 0.01V^2$$

$$0.01V^2 = 10 - 10e^{-0.2}$$

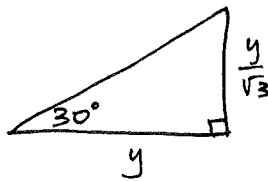
$$V = 13.46 \text{ m/s}$$

Academic Year		Calendar Year	
Course		Name of task/exam	

15a)

Method 1

Cross-section



$$A = \frac{1}{2} y \cdot \frac{y}{\sqrt{3}}$$

$$= \frac{y^2}{2\sqrt{3}}$$

$$V = \int_{-4}^4 \frac{y^2}{2\sqrt{3}} dx \quad (\text{Note thickness is on } x\text{-axis})$$

$$= \frac{1}{2\sqrt{3}} \int_{-4}^4 y^2 dx \quad \begin{matrix} x^2 + y^2 = 16 \\ y^2 = 16 - x^2 \end{matrix}$$

$$= \frac{1}{2\sqrt{3}} \int_{-4}^4 (16 - x^2) dx$$

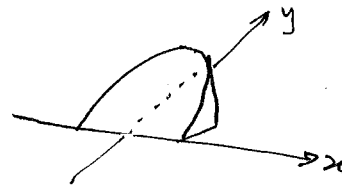
$$= \frac{2}{2\sqrt{3}} \int_0^4 (16 - x^2) dx$$

$$= \frac{1}{\sqrt{3}} \left[16x - \frac{x^3}{3} \right]_0^4$$

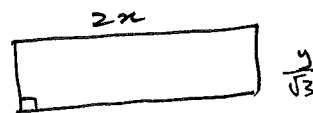
$$= \frac{1}{\sqrt{3}} \left[64 - \frac{64}{3} \right]$$

$$= \frac{128}{3\sqrt{3}}$$

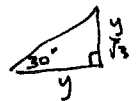
$$= \frac{128\sqrt{3}}{9} \text{ u}^3$$

Method 2

Cross-section



Aerial view



$$A = 2x \cdot \frac{y}{\sqrt{3}}$$

$$= \frac{2}{\sqrt{3}} xy$$

$$V = \int_0^4 \frac{2}{\sqrt{3}} xy dy \quad (\text{Note thickness on } y\text{-axis})$$

$$= \frac{2}{\sqrt{3}} \int_0^4 xy dy \quad \begin{matrix} x^2 + y^2 = 16 \\ x = (16 - y^2)^{1/2} \end{matrix}$$

$$= \frac{2}{\sqrt{3}} \int_0^4 (16 - y^2)^{1/2} \cdot y dy$$

$$\text{Note: } \frac{d}{dy} (16 - y^2)^{3/2} = \frac{3}{2} (16 - y^2)^{1/2} \cdot (-2y) = -3(16 - y^2)^{1/2}$$

$$= \frac{-2}{3\sqrt{3}} \left[(16 - y^2)^{3/2} \right]_0^4$$

$$= \frac{-2}{3\sqrt{3}} \left[0 - (16)^{3/2} \right]$$

$$= \frac{-2}{3\sqrt{3}} (-64)$$

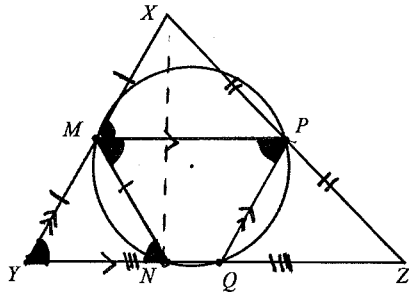
$$= \frac{128}{3\sqrt{3}}$$

$$= \frac{128\sqrt{3}}{9} \text{ u}^3$$

Solutions for exams and assessment tasks

Academic Year		Calendar Year	
Course		Name of task/exam	

b



i Since m is the midpoint of XY and P is the midpoint of XZ , then $MP \parallel YZ$ since the ratio of intercepts are equal.

Similarly, M is the midpoint of XY and Q is the midpoint of YZ , then $XY \parallel PQ$ since the ratio of intercepts are equal.

$\therefore MPQY$ is a parallelogram.

ii Since $MPQN$ is a cyclic quadrilateral, $\angle MNY = \angle MPQ$ (exterior angle in a cyclic quadrilateral equals the interior opposite angle.)

$$\therefore \angle MNY = \angle MPQ$$

iii since $MPQY$ is a parallelogram
 $\angle YNM = \angle NMP$ (alternate angles equal $MP \parallel QY$).

$$\text{Let } \angle MNY = x$$

$$\therefore \angle NMP = x$$

Since $\angle MNY = \angle MPQ$
 (proved in ii)

$$\text{then } \angle MPQ = x$$

Also $\angle MYN = x$ (opposite angles in parallelogram are equal).

$$\angle XMN = 2x$$

(exterior angle of triangle equals sum of 2 interior opposite angles).

$$\text{and since } \angle PMN = x$$

$$\text{then } \angle XMP = x \text{ also.}$$

$\therefore \triangle XMN$ is isosceles ($MY = MX$ and $MN = MX$)

$$\therefore \angle MNX = \frac{180 - 2x}{2} \quad (\text{angle sum of } \triangle XMN)$$

$$= 90 - x$$

$$\therefore \angle YNX = \angle YNM + \angle MNX$$

(adjacent angles)

$$= x + 90 - x$$

$$= 90$$

$$\therefore XN \perp YZ$$

Academic Year		Calendar Year	
Course		Name of task/exam	

c) $x = R \cos \frac{2\pi t}{T}$

↓
Σ
c

$$\dot{x} = -R \sin \frac{2\pi t}{T} \times \frac{2\pi}{T}$$

$$= -\frac{R 2\pi}{T} \sin \frac{2\pi t}{T}$$

$$\ddot{x} = -\frac{2\pi R}{T} \cos \frac{2\pi t}{T} \cdot \frac{2\pi}{T}$$

$$= -\frac{4\pi^2 R}{T^2} \cos \frac{2\pi t}{T}$$

$$= -\frac{4\pi^2}{T^2} \left(R \cos \frac{2\pi t}{T} \right)$$

$$\therefore \ddot{x} = -\frac{4\pi^2}{T^2} x$$

$$y = R \sin \frac{2\pi t}{T}$$

$$\dot{y} = R \cos \frac{2\pi t}{T} \times \frac{2\pi}{T}$$

$$= \frac{2\pi}{T} R \cos \frac{2\pi t}{T}$$

$$\ddot{y} = \frac{2\pi}{T} R \left(-\sin \frac{2\pi t}{T} \right) \left(\frac{2\pi}{T} \right)$$

$$= -\frac{4\pi^2}{T^2} \left(R \sin \frac{2\pi t}{T} \right)$$

$$\therefore \ddot{y} = -\frac{4\pi^2}{T^2} y$$

ii) $\text{accel} = \sqrt{\ddot{x}^2 + \ddot{y}^2}$

$$= \sqrt{\left(-\frac{4\pi^2}{T^2} \right)^2 x^2 + \left(-\frac{4\pi^2}{T^2} \right)^2 y^2}$$

$$= \sqrt{\frac{16\pi^4}{T^4} x^2 + \frac{16\pi^4}{T^4} y^2}$$

$$\text{accel} = \sqrt{\frac{16\pi^4}{T^4} (x^2 + y^2)}$$

as $x^2 + y^2 = R^2$ (P is on circle)

$$\therefore \text{accel} = \sqrt{\frac{16\pi^4}{T^4} R^2}$$

$$= \frac{4\pi^2}{T^2} R$$

$$= \frac{4\pi^2 R}{T^2}$$

as the accel is towards the centre of the circle

$$\text{accel} = -\frac{4\pi^2 R}{T^2}$$

iii) Force exerted by star on planet is the same as the force exerted by the planet on the star.

$$\therefore \text{Force} = m \left(\frac{4\pi^2 R}{T^2} \right)$$

iv) $F = \frac{GMm}{R^2}$

$$m \left(\frac{4\pi^2 R}{T^2} \right) = \frac{GMm}{R^2}$$

$$4\pi^2 R = T^2 \frac{GM}{R^2}$$

Solutions for exams and assessment tasks

Academic Year		Calendar Year	
Course		Name of task/exam	

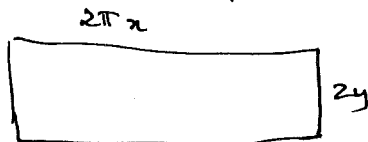
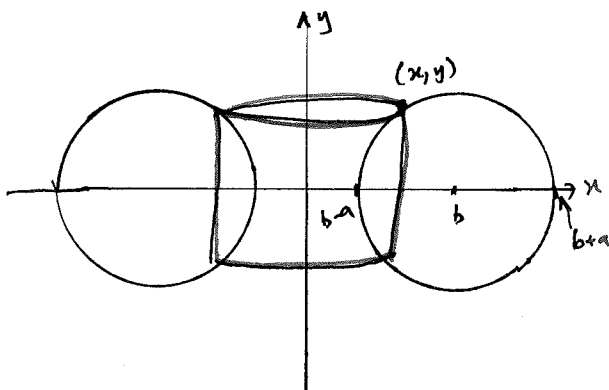
$$T^2 = \frac{4\pi^2 R^3}{GM}$$

$$\therefore T = 2\pi R \sqrt{\frac{R}{GM}}$$

d

with respect to x :

means dx so width needs to be measured in x so need to use cylindrical shells.



$$A = 4\pi xy$$

$$V = 4\pi \int_{b-a}^{b+a} xy \, dx$$

$$= (x-b)^2 + y^2 = a^2$$

$$y^2 = a^2 - (x-b)^2$$

$$y = \sqrt{a^2 - (x-b)^2}$$

$$\therefore V = 4\pi \int_{b-a}^{b+a} x \cdot \sqrt{a^2 - (x-b)^2} \, dx$$

Academic Year		Calendar Year	
Course		Name of task/exam	

16a(i)

$$I_n = \int_0^{\pi/2} \sin^n \theta \, d\theta$$

$$= \int_0^{\pi/2} \sin^{n-1} \theta \cdot \sin \theta \cdot d\theta$$

$$u = \sin^{n-1} \theta$$

$$du = (n-1) \sin^{n-2} \theta \cdot \cos \theta \, d\theta$$

$$v = \cos \theta$$

$$v' = -\sin \theta$$

$$\therefore I_n = \left[-\sin^{n-1} \theta \cdot \cos \theta \right]_0^{\pi/2} +$$

$$(n-1) \int_0^{\pi/2} \sin^{n-2} \theta \cdot \cos^2 \theta \, d\theta$$

$$= 0 +$$

$$(n-1) \int_0^{\pi/2} \sin^{n-2} \theta \cdot (1 - \sin^2 \theta) \, d\theta$$

$$I_n = (n-1) \int_0^{\pi/2} \sin^{n-2} \theta \, d\theta -$$

$$(n-1) \int_0^{\pi/2} \sin^n \theta \, d\theta$$

$$I_n = (n-1) I_{n-2} - (n-1) I_n$$

$$I_n + (n-1) I_n = (n-1) I_{n-2}$$

$$I_n (1 + n - 1) = (n-1) I_{n-2}$$

$$n I_n = (n-1) I_{n-2}$$

$$I_n = \frac{n-1}{n} I_{n-2}$$

(ii)

$$\int_0^2 (4-x^2)^{\frac{5}{2}} \, dx$$

$$x = 2 \cos \theta$$

$$dx = -2 \sin \theta \, d\theta$$

$$x=0 \quad x=2$$

$$\theta = \frac{\pi}{2} \quad \theta = 0$$

$$= \int_{\pi/2}^0 -(4-4\cos^2 \theta)^{\frac{5}{2}} \cdot 2 \sin \theta \, d\theta$$

$$= \int_0^{\pi/2} (4\sin^2 \theta)^{\frac{5}{2}} \cdot 2 \sin \theta \, d\theta$$

$$= 4 \cdot 2 \int_0^{\pi/2} \sin^5 \theta \cdot \sin \theta \, d\theta$$

$$= 2 \cdot 2 \int_0^{\pi/2} \sin^6 \theta \, d\theta$$

$$= 2^6 \int_0^{\pi/2} \sin^6 \theta \, d\theta$$

$$\int_0^{\pi/2} \sin^6 \theta \, d\theta = I_6$$

$$= \frac{5}{6} I_4$$

$$= \frac{5}{6} \cdot \frac{3}{4} \cdot I_2$$

$$= \frac{5}{6} \times \frac{3}{4} \times \frac{1}{2} I_0$$

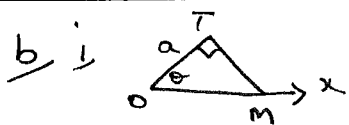
$$= \frac{5}{16} \int_0^{\pi/2} 1 \, d\theta$$

$$= \frac{5}{16} [\theta]_0^{\pi/2}$$

$$= \frac{5}{16} \cdot \frac{\pi}{2}$$

$$\therefore \int_0^2 (4-x^2)^{\frac{5}{2}} \, dx = 64 \cdot \frac{5}{16} \cdot \frac{\pi}{2} = 10\pi$$

Academic Year		Calendar Year	
Course		Name of task/exam	



$$\text{since } x^2 + y^2 = a^2$$

$$\text{radius} = a$$

$$\therefore OT = a$$

$$\cos \theta = \frac{a}{OM}$$

$$OM = \frac{a}{\cos \theta}$$

$$= a \sec \theta$$

$\therefore$ coordinates of M are

$$(a \sec \theta, 0)$$

ii P has the same x value as M

$\therefore$ x value is $a \sec \theta$.

The y-value lies on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

$$\therefore \frac{(a \sec \theta)^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\frac{a^2 \sec^2 \theta}{a^2} - \frac{y^2}{b^2} = 1$$

$$\sec^2 \theta - 1 = \frac{y^2}{b^2}$$

$$\tan^2 \theta = \frac{y^2}{b^2}$$

$$y^2 = b^2 \tan^2 \theta$$

$y = b \tan \theta$ in first quad.

$$\therefore P(a \sec \theta, b \tan \theta)$$

$$\text{iii } m_{PQ} = \frac{b \tan \theta - b \tan \phi}{a \sec \theta - a \sec \phi}$$

eqn chord PQ:

$$y - b \tan \theta = \frac{b \tan \theta - b \tan \phi}{a \sec \theta - a \sec \phi} (x - a \sec \theta)$$

$$y = \frac{b \tan \theta - b \tan \phi}{a \sec \theta - a \sec \phi} (x - a \sec \theta) + b \tan \theta$$

$$\phi = \frac{\pi}{2} - \theta$$

$$y = \frac{b \tan \theta - b \tan(\frac{\pi}{2} - \theta)}{a \sec \theta - a \sec(\frac{\pi}{2} - \theta)} [x - a \sec \theta] + b \tan \theta$$

$$= \frac{b \tan \theta - b \cot \theta}{a \sec \theta - a \operatorname{cosec} \theta} [x - a \sec \theta] + b \tan \theta$$

$$= \frac{b(\tan \theta - \cot \theta)}{a(\sec \theta - \operatorname{cosec} \theta)} [x - a \sec \theta] + b \tan \theta$$

$$= \frac{b}{a} \left(\frac{\frac{\sin \theta}{\cos \theta} - \frac{\cos \theta}{\sin \theta}}{\frac{1}{\cos \theta} - \frac{1}{\sin \theta}} \right) [x - a \sec \theta] + b \tan \theta$$

$$= \frac{b}{a} \left[\frac{\sin^2 \theta - \cos^2 \theta}{\sin \theta - \cos \theta} \right] [x - a \sec \theta] + b \tan \theta$$

$$= \frac{b}{a} \left(\frac{\sin \theta - \cos \theta}{\sin \theta - \cos \theta} (\sin \theta + \cos \theta) \right) [x - a \sec \theta] + b \tan \theta$$

$$= \frac{b}{a} (\sin \theta + \cos \theta) (x - a \sec \theta) + b \tan \theta$$

Page of

Academic Year		Calendar Year	
Course		Name of task/exam	

$$y = \frac{b}{a} (\sin \theta + \cos \theta)x - \frac{b}{a} (\sin \theta + \cos \theta) \frac{a \sec \theta}{\cos \theta} + b \tan \theta$$

$$= \frac{b}{a} (\cos \theta + \sin \theta)x - \frac{b}{a} (\sin \theta + \cos \theta) \frac{a}{\cos \theta} + b \frac{\sin \theta}{\cos \theta}$$

$$= \frac{b}{a} (\cos \theta + \sin \theta)x - \frac{b}{a} \frac{a \sin \theta}{\cos \theta} - \frac{b}{a} \frac{\cos \theta a}{\cos \theta} + b \frac{\sin \theta}{\cos \theta}$$

$$= \frac{b}{a} (\cos \theta + \sin \theta)x - \frac{b \sin \theta}{\cos \theta} - \frac{b}{a} a + b \frac{\sin \theta}{\cos \theta}$$

$$\therefore y = \frac{b}{a} (\cos \theta + \sin \theta)x - b$$

iv Every chord has an equation $y = mx + b$; and every chord must pass through the y-intercept of $-b$
 $\therefore$ the fixed point is $(0, -b)$

$$v \quad y = \pm \frac{b}{a} x$$

$$vi \quad \text{as } \theta \rightarrow \frac{\pi}{2}$$

$$y \rightarrow \frac{b}{a} (0 + 1)x - b$$

$$\therefore y = \frac{b}{a} x - b$$

which is parallel to the asymptote $y = \frac{b}{a} x$ since $m_1 = m_2$ for parallel lines.

$\therefore$ PQ approaches ~~the~~ a line parallel to an asymptote.